CL 299 presentation 3

LNP Manufacturing and Process Simulation

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PROCESS FLOW CHART –LAB SCALE

- We made the flow chart using draw.io software.
- In this flow chart we have covered mixing, TFF, ultra filtration, quenching, sterile filtration and cold storage.
- For aqueous phase we have used 25mM citrate buffer with target pH~4.
- For both the aqueous and organic phases, the temperature is between 20 – 25°C.
- we are using macrofluidic mixing as it is easier to operate at lab scale.

Alternatives of M101(macrofluidic mixture)

1. Microfluidic mixing.	2.Static T-Mixer	3.Rapid Injection / Vortex Mixing
-Hydrodynamic focusing -Controlled laminar diffusion mixing	(Syringe Pump-Based)	
* Precise FRR and TFR control * Low PDI (<0.2) * High reproducibility * Suitable for 0.5–50 mL batches	* Two syringe pumps * T-junction connector * Controlled FRR (e.g., 3:1) * Total flow 2–12 mL/min	* Inject lipid ethanol into stirring mRNA buffer * Magnetic stirrer or vortex
Best for research optimization *Precise size control*	*Low cost* *Easy to assemble* *Slightly higher size variability*	*Simple setup* *No pump required* *Less size control* *Higher PDI possible*

Alternatives for UF101 (Ethanol Removal & Buffer Exchange)

1. Amicon Ultra Centrifugal Filters (Most Common)	2.Dialysis Tubing System	3.Stirred Cell Ultrafiltration (Pressure-Driven)
* 100–300 kDa MWCO * Spin at 3,000–5,000 g * Volume: 1–50 mL * Used for: Ethanol removal Buffer exchange Concentration *Fast* *Widely used in labs* *Simple operation*	* 100–300 kDa membrane * Buffer bath under stirring * 4–12 hour exchange *Very low cost* *No pump required* *Slow process* *Less control over TMP*	* Nitrogen pressure system * Controlled membrane filtration * Better scalability than centrifuge filters *More professional setup* *Better process control* *Suitable for 50–200 mL*

Alternatives for F101 (Sterile Filtration)

1. 0.22 µm Syringe Filter (Standard Lab Method)	2. Vacuum Filtration Assembly	3. Sterile Single-Use Assembly (Advanced Lab Setup)
* PES / PVDF membrane * Manual sterile filtration * Used inside biosafety cabinet	* 0.22 µm membrane * Vacuum pump * For 50–200 mL batches	* Pre-sterilized tubing * Inline 0.22 µm filter * Closed transfer system
Most common method *Easy and affordable* *Suitable for small batches*	*Faster for larger volumes* *Controlled sterile handling*	*Mimics GMP process* *Reduces contamination risk* *Suitable for translational research*

Catalogue Code

- We have used python as our coding language using the Google Colab for writing and executing code.
- We took reference of some patents and extracted the data from these patents using AI tools.
- We extracted information about various components of LNPs (particularly ionizable lipids, helper lipids, PEG-lipids and sterols).
- In data we have list of various compounds for each components of LNPs and their molecular weights.
- <u>There are many compounds for which we have a range of molecular weight because they have number of units (n) which varies across mixture.</u>
- Also, we have distribution of various categories.

```python
from lnp_molecular_library import create_lnp_library

lnp_lib = create_lnp_library()

print(lnp_lib.search_by_category("Ionizable Lipid"))
```

```
[DLin-MC3-DMA (Ionizable Lipid) → 642.1 g/mol, DLin-KC2-DMA (Ionizable Lipid) → 638.0 g/mol, DLin-DMA (Ionizable
```

```python
part = lnp_lib.search_by_name("Lathosterol")

print(f"LNP Part: {part.name}")
print(f"Category: {part.category}")
print(f"Molecular Weight: {part.get_molecular_weight()}")
```

```
LNP Part: Lathosterol
Category: Sterol
Molecular Weight: 386.65 g/mol
```

```
[40]
from lnp_molecular_library import create_lnp_library

lnp_lib = create_lnp_library()

print(lnp_lib.search_by_name("SM-102"))
print(lnp_lib.search_by_category("Ionizable Lipid"))
```

```
SM-102 (Ionizable Lipid) → 766.3 g/mol
[DLin-MC3-DMA (Ionizable Lipid) → 642.1 g/mol, DLin-KC2-DMA (Ionizable Lipid) → 638.0 g/mol, DLin-DMA (Ionizable
```

PAT Measurement Level

- describe how closely an analytical method is connected to the manufacturing process and how useful it is for real-time control.
- Four Analytical levels:

  Offline: analysis after sample removal.

  At-line: analysis near the manufacturing line.

  Online: automated sampling and analysis loop.

  In-line: sensor placed directly in the process stream.

  Process relevance and control capability increase from offline to Inline.

Offline Analysis

- A sample is removed from process.
- Transported to a separate laboratory.
- Then analysed and results are obtained.
- Perform by manual sampling that can cause some sampling errors.
- Not real time monitoring.
- High accuracy and advance analysis.

Offline measurement is the traditional approach where quality is checked after sampling instead of during processing.

At-line Analysis

- A sample is removed from the process.
- Analysed using equipment located near the production line.
- Faster analysis compared to offline laboratory testing.
- Performed by manual sampling and sampling errors may still occur.
- Not fully real-time monitoring but provides quicker feedback.
- Allows rapid process monitoring and adjustment.

At-line measurement is an intermediate approach where analysis is done close to the process, providing faster results than offline analysis but still requiring sample removal.

Online Analysis

- Sample is automatically taken from the process using a bypass system.
- Analytical equipment is connected directly to the production line.
- Sample may be conditioned before analysis (e.g., dilution or temperature control).
- Provides near real-time monitoring.
- Reduces manual sampling errors.
- Allows continuous process monitoring and faster feedback.

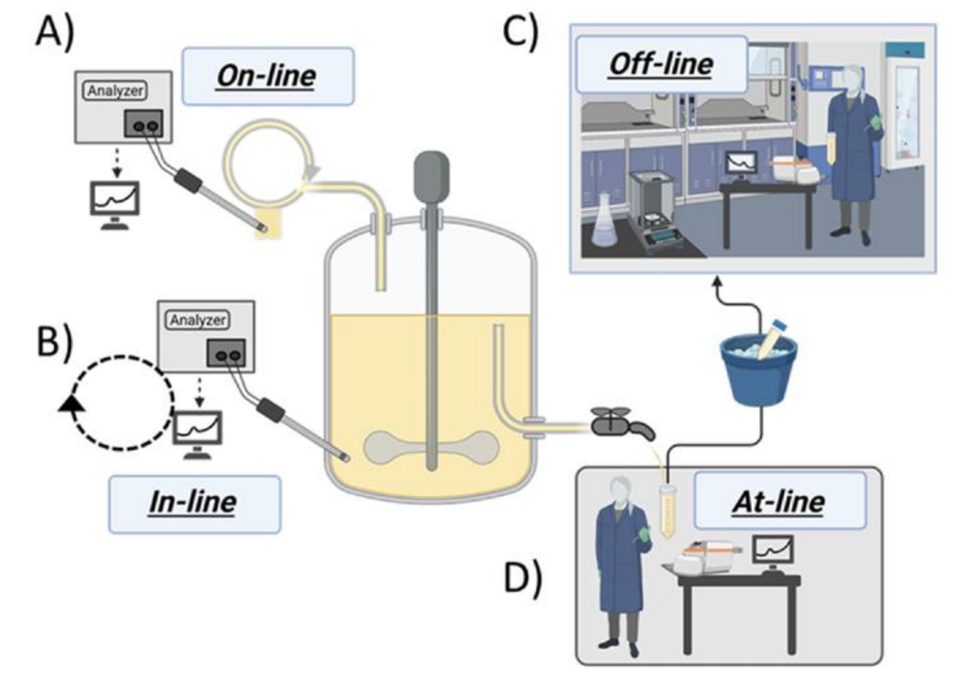
The disadvantages of online analysis are the higher capital costs that arise from the automation and the requirements for robustness of the analytical equipment.

In-line Analysis

- Measurement is performed directly inside the process.
- Analytical sensor is integrated into the production line.
- No sample removal is required.
- Provides continuous and real-time monitoring.
- Enables immediate detection of process changes.
- Supports automatic process control and optimization.

Inline measurement is an in-situ approach where analysis occurs directly within the process stream, allowing real-time monitoring and control without manual sampling.

Level	Sample removal	Real-time	Control capability
Offline	Yes	No	Low
At-line	Yes	Limited	Medium
Online	Automated	Near real-time	High
Inline	No	Real-time	Highest



CLASSIFICATION OF TECHNIQUES

- **Near-infrared spectroscopy (NIR)**
  Measures molecular vibrations.
  Used for real-time process monitoring as online and inline.

- **Infrared spectroscopy**
  This technique is used to identify chemical bonds and molecular structures
  Used for real-time process monitoring as online and inline(major).

- **Nuclear magnetic resonance (NMR)**
  This technique is used to determine molecular structure and chemical composition using magnetic properties of atomic nuclei.
  Offline and online.

- **Raman spectroscopy**
  This technique is used to analyze molecular structure and composition by measuring light scattering from molecules
  online and inline.

- **Dynamic light scattering (DLS)**
  This technique is used to measure particle size and size distribution in real-time.
  Inline.

- **Fluorescence**
  This technique is used to detect specific molecules and monitor nanoparticle formation or payload distribution
  offline

- **Surface plasmon resonance (SPR)**
  Surface plasmon resonance is an optical sensing technique that measures changes in refractive index at a metal surface to detect molecular interaction.
  Inline.

- **reflective interference spectroscopy (RIF)**
  measures interference patterns of reflected light to detect changes in surface thickness or refractive index.
  Inline.

- **Mass spectrometry (MS)**
  provides information about the molecular weight and composition of lipids
  online.

- **Electrochemical sensors**
  measure the chemical properties.
  Inline.

- **NIR Turbidity**
  uses Near-Infrared (NIR) light to determine how cloudy or particle-rich a liquid is.
  Inline.

- **Capacitance**: Inline.

- **IR Absorption**
  Infrared (IR) absorption occurs when molecules absorb infrared radiation, causing changes in vibrational and rotational energy levels.
  Online and inline.

- **(Bio)chemical Analysers**
  used to measure chemical or biological components
  offline.

- **Calorimetry**
  measuring heat transfer during physical or chemical processes
  online.

- **Confocal Spectroscopy (CICS)**
  used for high-resolution nanoparticle characterization
  offline.

- **Consistent cryo-electron microscopy (Cryo-EM)**
  This technique is used to visualize nanoparticle structure and morphology at very high resolution.
  offline

- **Small-angle X-ray scattering (SAXS).**
  This technique is used to analyze nanoscale structure and internal organization of particles using X-ray scattering.
  offline

- **Computational Fluid Dynamics (CFD)**
  uses mathematical modeling and simulations.
  Offline.

- **Zeta potential Analysis**
  This technique determines the surface charge.
  Inline and at-line.

- **Optical Microscopy and Electron Microscopy**
  helping to identify any issues like agglomeration or structural defects.
  Offline.

- **Differential Scanning Calorimetry (DSC)**
  (Thermal analysis, not real-time).
  Offline.

- **X-ray Diffraction (XRD)**
  This technique is used to determine crystal structure and phase composition by analyzing X-ray diffraction patterns.
  Offline.

- **High-Performance Liquid Chromatography (HPLC)**
  On-line possible via auto-samplers for drug loading, encapsulation efficiency, or degradation monitoring.

- **Fourier Transform Infrared Spectroscopy (FTIR)**
  Can be on-line with ATR-FTIR probes, but typically offline unless specialized probe systems are used.

- **UV spectroscopy**
  used to measure chemical changes and molecular concentration.

- **Ultrasonic crystallization monitoring (UCM)**
  uses ultrasonic waves to monitor crystallization processes by detecting particle formation and phase changes

- **Photometric stereo imaging**
  Photometric stereo imaging reconstructs surface shape by analyzing images taken under different lighting angles.
  At-line and online.

- **FBRM (Focused Beam Reflectance Measurement)**
  this technique is used to monitor particle size and count in real time

- **PVM (Particle Vision and Measurement)**
  captures real-time images of particles inside a process to monitor particle shape, size, and behavior.
  Inline.

Resources

- https://docs.google.com/presentation/d/18YQe9UQSGdCi0CvTygbpCGwfLPhk4ogJ/edit?usp=sharing&ouid=114130096820631242211&rtpof=true&sd=true
- https://drive.google.com/file/d/1qwG77msmF2oO8doUGwgtkrXdz0KD0xtk/view?usp=sharing

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THANK YOU!!!